



## GRAHAM PACKAGING: SAVING ENERGY THROUGH AUTOMATED IDLE CONTROL

### SECTOR

Industrial

### LOCATION

Modesto

### FINANCIAL OVERVIEW

\$75,000

### BACKGROUND

Graham Packaging produces sustainable polyethylene terephthalate (PET) and high-density polyethylene (HDPE) plastic bottles for the food, beverage, home care, and industrial industries. At its Modesto, CA plant, Graham implemented a project to automatically start-and-stop idle electric and compressed air loads to improve energy usage. The plant achieved a 3% reduction, approximately 500,000 kWh, in electricity usage during the first year by controlling motors and compressed air systems that had previously functioned continuously while not loaded and fully operational. Managing these systems more effectively reduced annual energy costs by \$48,000 and optimized the equipment's operation for higher performance. The success of this project allowed Graham to invest in this technology at its other plants, contributing to the company's goal of reducing GHG emissions and energy usage.

### SOLUTIONS

Working in partnership with an external vendor and internal technical team, Graham created and implemented a vigorous timeline from conception to completion, while reducing the risk of project changes and staying within budget. The project cost \$75,000 and was completed in two months, resulting in energy savings of \$48,000 annually with a simple payback of 1.6 years.

The team was tasked with improving operational efficiencies through an assessment and in-depth audit of the Joplin facility. The audit was completed using sensors and portable meters borrowed for free through the DOE [Better Plant Program's Diagnostic Equipment Program](#). Through this program, an expert from the Oak Ridge National Laboratory worked with facility managers to select appropriate meters and sensors, ensured they were set up correctly, and analyzed the data on machine downtime and kilowatt-hour usage. Projected energy savings were calculated after months of data logging and analysis. The data indicated that significant energy savings were achievable through power usage controls.

However, the project was not without challenges, including the need to ensure the machine's automatic start-and-stop sequence did not disrupt line operations or create other operational issues. Graham also needed to limit the downtime of equipment during implementation to continue to meet customer demand. The team overcame these challenges by building the project control panel off-site and combining the developed project timeline with other projects that were completed during scheduled shutdowns of the facility. This lessened the impact on production and reduced disruptions.

To stay within budget, the maintenance staff at the facility assisted in the installation of the controls system and were thoroughly trained in the operation, preventative maintenance, and limitations of the new system. This prepared Graham staff to continue to provide the proper care and maintenance of the equipment and ensure continued performance. After installation, a program was built to track every time the line was shut down and sum the total shutdown time. The time was then multiplied by the kilowatt loads avoided by the system during idle time. Actual energy savings were recorded and tracked using remote access to the management system. Through this process, the plant saw a 3% reduction in electricity usage sitewide (approximately 500,000 kWh) during the first year of the project.

## OTHER BENEFITS

Due to the success at the Modesto facility, Graham is looking to deploy similar energy savings projects at other plants throughout the United States. While this technology is not a one-size-fits-all solution for eliminating excessive energy usage, the successful implementation at this plant is scalable for others to follow. It will continue to be shared and promoted as a best practice to help Graham achieve its goal of a 25% energy reduction by 2028.